10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 5.3 — Multiplication & Division

Unit 5 | Solved Exercise | New Book (PECTAA)

Exercise 5.3 (Solved)

For multiplication we factorise, cancel the common factors, and then multiply what is left. For division we change ÷ into × and turn the second fraction upside down (write its reciprocal).

1. Simplify.

(i) $\frac{24lm}{5l + 10m} \times \frac{5l^2 - 20m^2}{16mn}$

Take out the common factors: $5l + 10m = 5(l + 2m)$ and $5l^2 - 20m^2 = 5(l - 2m)(l + 2m)$

$\frac{24lm}{5(l + 2m)} \times \frac{5(l - 2m)(l + 2m)}{16mn}$

$= \frac{24lm(l - 2m)}{16mn}$

Divide 24 and 16 by 8

$= \frac{3lm(l - 2m)}{2mn}$

Cancel m

$= \frac{3l(l - 2m)}{2n}$

Answer: $3l(l - 2m)/(2n)$

(ii) $\frac{x^2 - 3x + 2}{x^2 + 3x - 4} \times \frac{2x^2 + 8x}{3x + 6}$

Factorise every expression: $(x - 1)(x - 2)$, $(x + 4)(x - 1)$, $2x(x + 4)$ and $3(x + 2)$

$\frac{(x - 1)(x - 2)}{(x + 4)(x - 1)} \times \frac{2x(x + 4)}{3(x + 2)}$

$= \frac{(x - 2)}{(x + 4)} \times \frac{2x(x + 4)}{3(x + 2)}$

Cancel $(x - 1)$ first, then cancel $(x + 4)$

$= \frac{2x(x - 2)}{3(x + 2)}$

Answer: $2x(x - 2)/[3(x + 2)]$

(iii) $\frac{a^2 - 4b^2}{a^2 + 2ba} \times \frac{2a^2 + 10ab}{a^2 + 3ab - 10b^2}$

Factorise: $(a - 2b)(a + 2b)$, $a(a + 2b)$, $2a(a + 5b)$ and $(a + 5b)(a - 2b)$

$\frac{(a - 2b)(a + 2b)}{a(a + 2b)} \times \frac{2a(a + 5b)}{(a + 5b)(a - 2b)}$

$= \frac{(a - 2b)}{a} \times \frac{2a}{(a - 2b)}$

Cancel $(a + 2b)$ and $(a + 5b)$, then cancel $(a - 2b)$

$= 2$

Answer: 2

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(iv) $\frac{(a+b)^2-c^2}{(a+c)^2-b^2} \times \frac{a^2-(b-c)^2}{(a+c)^2-c^2}$

Write every expression as a difference of two squares

$(a+b)^2-c^2=(a+b-c)(a+b+c)$

$(a+c)^2-b^2=(a-b+c)(a+b+c)$

$a^2-(b-c)^2=(a-b+c)(a+b-c)$

$(a+c)^2-c^2=a(a+2c)$

$= \frac{(a+b-c)(a+b+c)}{(a-b+c)(a+b+c)} \times \frac{(a-b+c)(a+b-c)}{a(a+2c)}$

$= \frac{(a+b-c)}{(a-b+c)} \times \frac{(a-b+c)(a+b-c)}{a(a+2c)}$

Cancel $(a+b+c)$ and then $(a-b+c)$

$= \frac{(a+b-c)^2}{a(a+2c)}$

Answer: $(a+b-c)^2/[a(a+2c)]$

(v) $\frac{x^3-8}{x^2-4} \div \frac{x^2+2x+4}{x^2+4x+4}$

Change ÷ into × and write the reciprocal of the second fraction

$= \frac{x^3-8}{x^2-4} \times \frac{x^2+4x+4}{x^2+2x+4}$

Now factorise the three expressions

$x^3-8=(x-2)(x^2+2x+4)$

$x^2-4=(x-2)(x+2)$

$x^2+4x+4=(x+2)^2$

$= \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)} \times \frac{(x+2)^2}{x^2+2x+4}$

$= \frac{(x+2)^2}{x+2}$

Cancel $(x-2)$, (x^2+2x+4) and one $(x+2)$

$= x+2$

Answer: x + 2

(vi) $((a^2+ab)/(a^2-ab))^2 \div ((a+b)/(a-b))^2$

Simplify inside the first bracket: $a^2+ab=a(a+b)$ and $a^2-ab=a(a-b)$

$= \left(\frac{(a+b)}{(a-b)}\right)^2 \div \left(\frac{(a+b)}{(a-b)}\right)^2$

Both the fractions are the same, so they cancel

$= 1$

Answer: 1

(vii) $\frac{x^2-x-6}{x^2-x-20} \div \left\{\frac{x^3-3x^2}{x^2+4x} \times \frac{x^3-5x^2-14x}{x^2-12x+35}\right\}$

First solve the bracket. Take out the common factors

$\frac{x^3-3x^2}{x^2+4x}$

$= \frac{x^2(x-3)}{x(x+4)}$

Cancel x

$= \frac{x(x-3)}{x+4}$

$\frac{x^3-5x^2-14x}{x^2-12x+35}$

$= \frac{x(x-7)(x+2)}{(x-5)(x-7)}$

Cancel $(x-7)$

$= \frac{x(x+2)}{x-5}$

So the bracket $= \frac{x^2(x-3)(x+2)}{(x+4)(x-5)}$

Now change ÷ into × and write the reciprocal

$= \frac{(x-3)(x+2)}{(x-5)(x+4)} \times \frac{(x+4)(x-5)}{x^2(x-3)(x+2)}$

Cancel all the common factors

$= \frac{1}{x^2}$

Answer: $1/x^2$

(viii) $\frac{3x^2-3xy}{10x^2+10xy-20y^2} \times \frac{5xy+10y^2}{6x^2+6xy+6y^2} \div \frac{2x^2-2y^2}{4x^3-4y^3}$

Factorise each expression

$= \frac{3x(x-y)}{10(x+2y)(x-y)} \times \frac{5y(x+2y)}{6(x^2+xy+y^2)} \div \frac{2(x-y)(x+y)}{4(x-y)(x^2+xy+y^2)}$

Cancel $(x-y)$ and write the last fraction in the simplest form

$= \frac{3x}{10(x+2y)} \times \frac{5y(x+2y)}{6(x^2+xy+y^2)} \times \frac{(x^2+xy+y^2)}{(x+y)}$

Cancel $(x-y)$ and (x^2+xy+y^2) inside, then change ÷ into ×

$= \frac{3x \times 5y \times 2}{10 \times 6 \times 2(x+y)}$

Cancel the numbers 6 and 30

$= \frac{xy}{2(x+y)}$

Answer: $xy/[2(x+y)]$

(ix) $\frac{2x^2-98}{x^3-125} \times \frac{x^2-3x-10}{3x-21} \div \frac{x^2+5x-14}{x^2+5x+25}$

Factorise every expression

$2x^2-98 = 2(x-7)(x+7)$ and $x^3-125 = (x-5)(x^2+5x+25)$

$x^2-3x-10 = (x-5)(x+2)$ and $3x-21 = 3(x-7)$

$x^2+5x-14 = (x+7)(x-2)$

$= \frac{2(x-7)(x+7)}{(x-5)(x^2+5x+25)} \times \frac{(x-5)(x+2)}{3(x-7)} \times \frac{x^2+5x+25}{(x+7)(x-2)}$

Change ÷ into × and cancel $(x-7)$, $(x-5)$, $(x+7)$ and $(x^2+5x+25)$

$= \frac{2(x+2)}{3(x-2)}$

Answer: $2(x+2)/[3(x-2)]$

(x) $\frac{x^2-2x}{2x+6} \times \frac{x^2+x-6}{x^2-5x} \times \frac{6x-30}{x^2-2x}$

Factorise: $x(x-2)$, $2(x+3)$, $(x+3)(x-2)$, $x(x-5)$, $6(x-5)$, $x(x-2)$

$= \frac{x(x-2)}{2(x+3)} \times \frac{(x+3)(x-2)}{x(x-5)} \times \frac{6(x-5)}{x(x-2)}$

Cancel $(x+3)$, x, $(x-5)$ and $(x-2)$ one by one

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$= \frac{6(x-2)}{2x}$

Divide 6 and 2 by 2

$= \frac{3(x-2)}{x}$

Answer: 3(x − 2)/x

Remember — multiplying and dividing fractions

1. a/b × c/d = ac/bd — factorise first, then cancel.
2. a/b ÷ c/d = a/b × d/c — turn the second fraction upside down.
3. Cancel one factor of the numerator with one equal factor of the denominator; never cancel across + or −.
4. $x^3 - y^3 = (x-y)(x^2 + xy + y^2)$ is used again and again in this exercise.

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